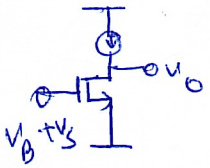
* Cascode Amplifier:

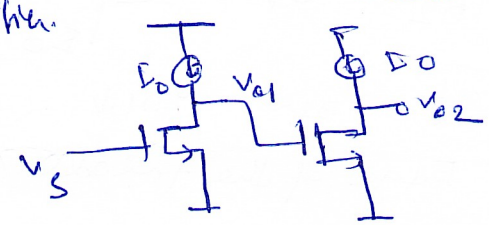


We know o/p voltage $V_o = -g_m r_{ds} \cdot V_S$ ideal current source

CS amplifier.

intrinsic gain is the largest gain.

Now we want more gain



$$V_{o1} = -g_m r_{ds} \cdot V_S$$

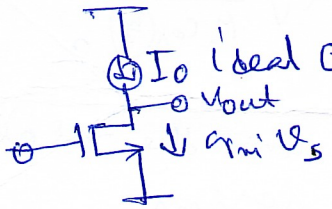
$$V_{o2} = -g_m r_{ds} \cdot V_{o1}$$

$$= -g_m r_{ds} \cdot (-g_m r_{ds}) V_S$$

$$= (g_m r_{ds})^2 \cdot V_S$$

$2 I_D \times V_{DD}$ is the DC Power consumed.

Different ~~cunt~~ element.



$$V_o = -g_m r_{ds} \cdot V_S$$

$\uparrow g_m$ or $r_{ds} \uparrow$

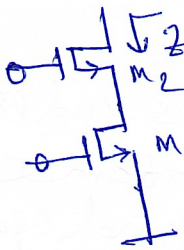
How do we increase

$\uparrow g_m$ by $\uparrow I_D$ and/or $\uparrow (W/L)$

How do we increase $\uparrow r_{ds}$ by $\uparrow L$ This is limited

This is limited

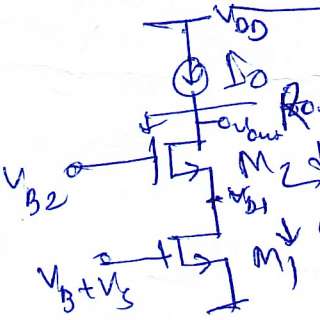
In other words



$Z_{out} = (g_{m2} r_{ds2}) r_{ds1}$

Cascode

cascode amplifier



$R_{out} = (g_{m2} r_{ds2}) \cdot r_{ds1}$

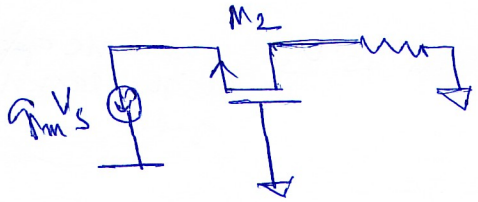
The output voltage

$V_o = -(g_{m1} V_S) \cdot R_{out}$

$$\frac{V_o}{V_S} = -g_{m1} (g_{m2} r_{ds2}) r_{ds1}$$

$$= -(g_{m1} r_{ds1}) (g_{m2} r_{ds2})$$

DC Power = $I_o V_{DD}$



M2 is simply a Common gate amplifier.

Cascod = CSA + CGA

$V_{B2} = V_{D1} + V_{GS2}\big|_{I_O}$

If $V_{B2}\downarrow$. $V_{D1}\downarrow$ Edge of the triode region

Minimum voltage possible

$V_{B2(min)} = V_{DSat\ of\ M_1} + V_{GS2}\big|_{I_O}$

$= \sqrt{\frac{2I_o}{\mu_n C_{ox}(\frac{W}{L})_1}} + V_{T2} + \sqrt{\frac{2I_o}{\mu_n C_{ox}(\frac{W}{L})_2}}$

$V_{B2(max)}$, $V_{B2}\uparrow$ $V_{D1}\uparrow$ M_2 go into the triode region.

$V_{B2(max - V_T)} = V_{ODC}$

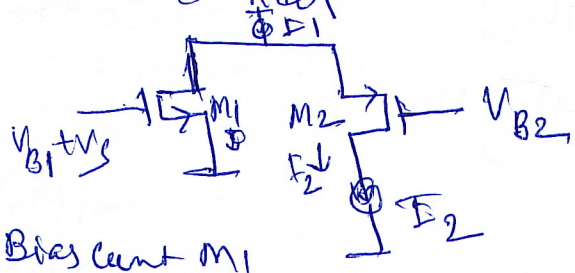
$V_{B2(max)} = V_{T2} + V_{ODC}$

Both transistors are in Saturation.

Cascode = CSA + CGA
↓ $\uparrow_{pmos}$ ↓ $\uparrow_{pmos}$
NMOS NMOS

Both NMOS and PMOS have the same Small Signal equivalent Circuit.

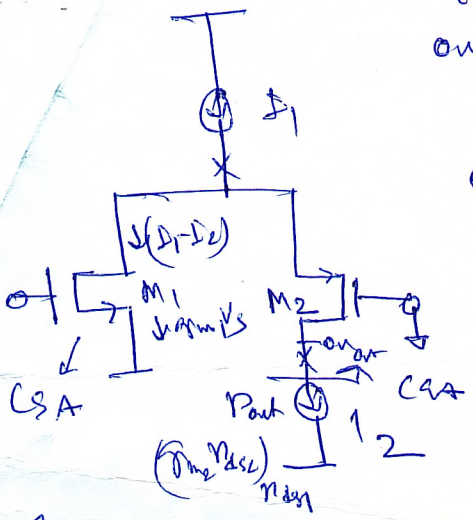
Let us consider



Bias current M_1 $(I_1 - I_2)$

Current flow in PMOS current flow S → D

Folded Cascode Amplifier.



Small-signal model.

overall output resistance

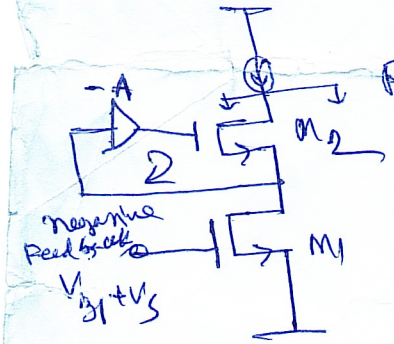
$$= (g_{m2} r_{ds2}) r_{ds1}$$

Overall transconductance g_{m1}

$$\frac{V_o}{V_s} = -(g_{m1} r_{ds1})(g_{m2} r_{ds2})$$

H.W Calculation of V_{B2} max and V_{B2} min.

How to increase the gain further?



$$R_{out} = A (g_{m2} r_{ds2}) r_{ds1}$$

Negative feedback pushes the output resistance

$$\frac{V_o}{V_s} = -(g_{m1} r_{ds1})(g_{m2} r_{ds2}) \cdot A$$

If $A = g_m r_{ds} \Rightarrow \frac{V_o}{V_s} \sim (g_m r_{ds})^3$

This particular circuit is called the "gain-boosted" cascode

